

BCSE204L Design and Analysis of Algorithms

Module-2: Topic-8: Job Sequencing problem

Dr Sunil Kumar P V

SCOPE

Job Selection/Sequencing Problem

- Given n jobs with profit, execution time, and deadline, achieve the schedule which maximizes the profit.
- Each job i is represented by the tuple (P_i, d_i, t_i) , where P_i , d_i , and t_i represent profit, deadline, and execution time associated with job i
- If job i is completed on or before its deadline, profit P_i is earned
- But if the job i finishes after its deadline, then a penalty P_i will incur.

Explanation by Example

- Maximization of profit can be redefined as minimization of penalty

Job	P_i	d_i	t_i
J_1	5	1	1
J_2	10	3	2
J_3	6	3	1
J_4	3	1	1

- Upper bound $\longrightarrow$ sum of all penalties except those are included in the solution
- $u(x) = \sum_{i \notin S_x} P_i$
- Cost $\longrightarrow$ sum of all penalties till the job under consideration
- $\hat{c}(x) = \sum_{i < m, i \notin S_x} P_i$ where $m = \max\{i / i \in S_x\}$
- S_x is the subset of jobs selected at node x

The procedure

- Demo: On board

Thank you...